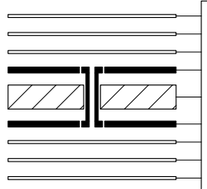
# DREAM FABRICATION DOCUMENT

## LAYER STACKUP LEGEND



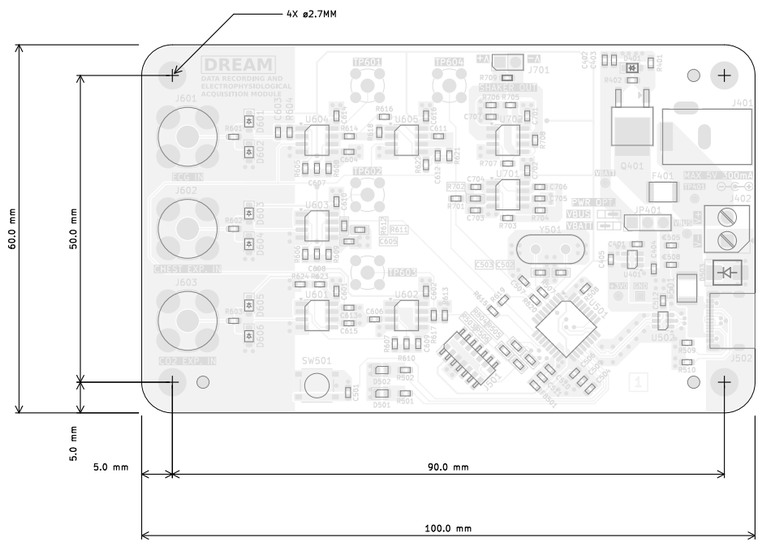
| Material | Layer | Thickness | Dielectric | Type | Gerber |
|---|---|---|---|---|---|
| | F.Paste | | | Paste Mask | |
| | F.Silkscreen | | Direct Printing | Legend | GBR |
| | F.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| **Copper** | **L1 (Sig, PWR)** | **0.035mm (1.00oz)** | | **Signal** | **GBR** |
| Core | | 1.49mm | FR4_7628 | Dielectric | |
| **Copper** | **L2 (Sig, GND)** | **0.035mm (1.00oz)** | | **Plane** | **GBR** |
| | B.Mask | 0.02mm | Solder Resist | Solder Mask | GBR |
| | B.Silkscreen | | Direct Printing | Legend | GBR |
| | B.Paste | | | Paste Mask | |

Total thickness: 1.6mm
Note: external layer thicknesses are specified after plating

## IMPEDANCE TABLE

| Transmission Line | Impedance [ohms] | Tolerance [ohms] | Layer | Trace Width [mm] | Gap [mm] | Ref. Layers |
|---|---|---|---|---|---|---|

## TOP FABRICATION (SCALE 1:1)




ALL DIMENSIONS IN MILLIMETERS UNLESS OTHERWISE SPECIFIED.

FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.
2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.
3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
4) SURFACE FINISH: IMMERSION GOLD
5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR GREEN.
6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.
7) ALL VIAS ARE TO BE NON-TENTED.
8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY RYAN DYNAMICS.
10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

| | |
|---|---|
| BOARD SIZE | 100.000 × 60.000 mm |
| BOARD THICKNESS | 1.600 mm |
| TRACE WIDTH | 0.200 mm |
| TRACE TO TRACE | 0.200 mm |
| MIN. HOLE (PTH) | 0.300 mm |
| MIN. HOLE (NPTH) | 0.650 mm |
| ANNULAR RING | 0.150 mm |
| COPPER TO HOLE | 0.254 mm |
| COPPER TO EDGE | 0.250 mm |
| HOLE TO HOLE | 0.254 mm |

| | | | |
|---|---|---|---|
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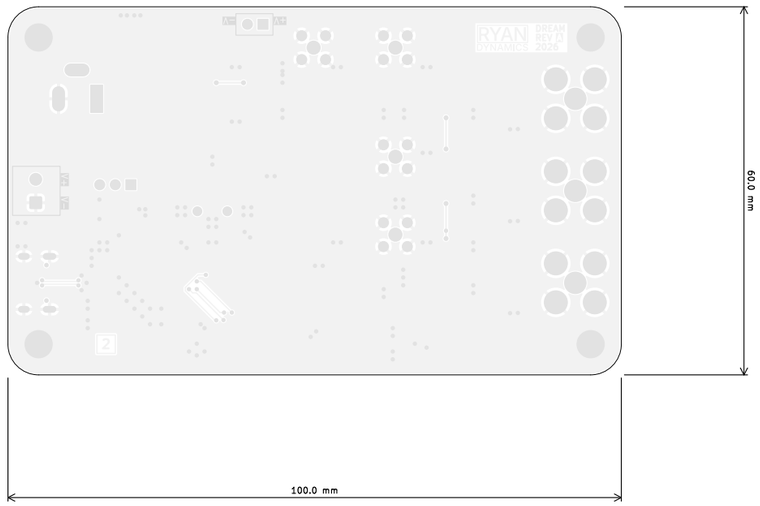
| CHK | | | |
| ENG APP | DREAM | | |
| MFR APP | | | |
| SHEET PATH | GIT HASH 53ebe71 | DRAWING No FABRICATION | A3 |
| FILENAME DREAM.kicad_pcb | VARIANT RELEASED | REVISION 1.0.1 | SHEET 1 OF 8 |

# DREAM FABRICATION DOCUMENT

## BOTTOM FABRICATION (SCALE 1:1)

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| ENG APP | DREAM | | |
| MFR APP | | | |
| SHEET PATH | GIT HASH 53ebe71 | DRAWING No FABRICATION | A3 |
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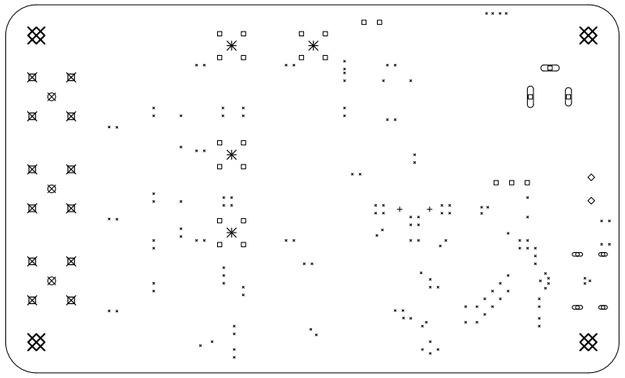
# DREAM FABRICATION DOCUMENT

## DRILL TABLE

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 145 | 0,30mm (11,81mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, GND) | Via |
| ○ | 4 | 0,60mm (23,62mils) | PTH | Slot | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| + | 2 | 0,80mm (31,50mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| □ | 21 + 3 | 1,00mm (39,37mils) | PTH | Round + Slot | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| ◇ | 2 | 1,10mm (43,31mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| ⊠ | 3 | 1,27mm (50,00mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| ✳ | 4 | 1,50mm (59,06mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| ⊠ | 12 | 1,52mm (60,00mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| ⊗ | 4 | 2,70mm (106,30mils) | PTH | Round | L1 (Sig, PWR) - L2 (Sig, GND) | Pad |
| | Total 200 | | | | | |

## DRILL DRAWING L1 – L2 (SCALE 1:1)



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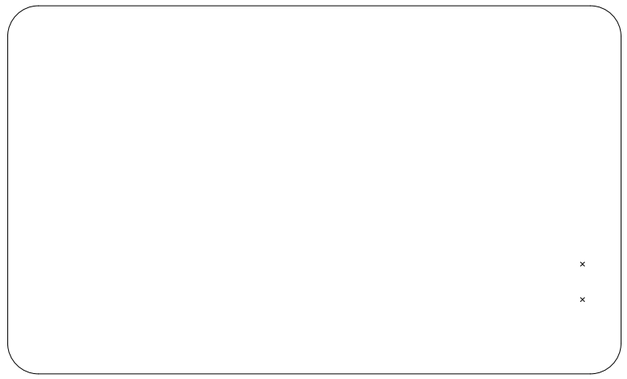
| FILENAME DREAM.kicad_pcb | VARIANT RELEASED | REVISION 1.0.1 | SHEET 3 OF 8 |

# DREAM FABRICATION DOCUMENT

## DRILL TABLE

| Symbol | Count | Hole Size | Plated | Hole Shape | Drill Layer Pair | Hole Type |
|---|---|---|---|---|---|---|
| × | 2 | 0.65mm (25.59mils) | NPTH | Round | L1 (Slg, PWR) - L2 (Slg, GND) | Mechanical |
| | Total 2 | | | | | |

## DRILL DRAWING L1 – L2 (SCALE 1:1)



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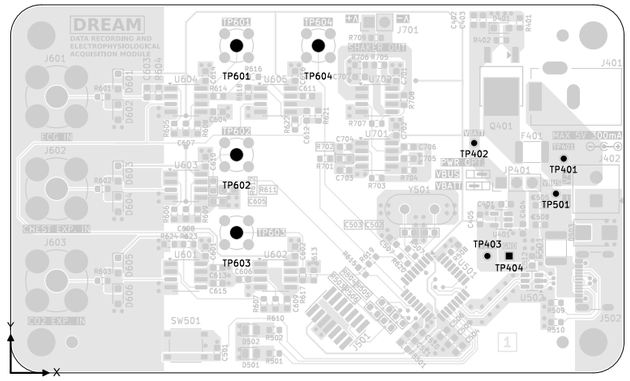
# DREAM FABRICATION DOCUMENT

## TOP TEST POINTS (SCALE 1:1)




| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|
| TP401 | VBATT_RAW | 90.17 | 34.92 |
| TP402 | VBATT | 75.56 | 37.46 |
| TP403 | +3V0 | 77.72 | 19.05 |
| TP404 | GND | 81.28 | 19.05 |
| TP501 | VBUS | 88.90 | 29.21 |
| TP601 | ECG_FILT,GND | 36.83 | 53.34 |
| TP602 | CE_FILT,GND | 36.83 | 35.56 |
| TP603 | CO2_OUT,GND | 36.83 | 22.86 |
| TP604 | ECG_OUT,GND | 50.16 | 53.34 |

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|

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| CHK | |
| ENG APP | DREAM |
| MFR APP | |

| SHEET PATH | GIT HASH | DRAWING No | |
|---|---|---|---|
| | 53ebe71 | FABRICATION | A3 |
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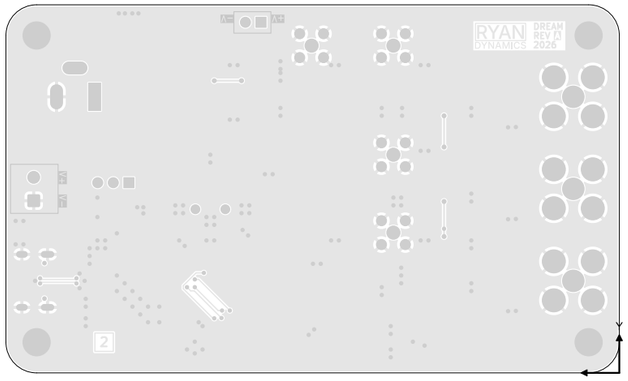
# DREAM FABRICATION DOCUMENT

## BOTTOM TEST POINTS (SCALE 1:1)

| Ref. | Net | X [mm] | Y [mm] |
|---|---|---|---|

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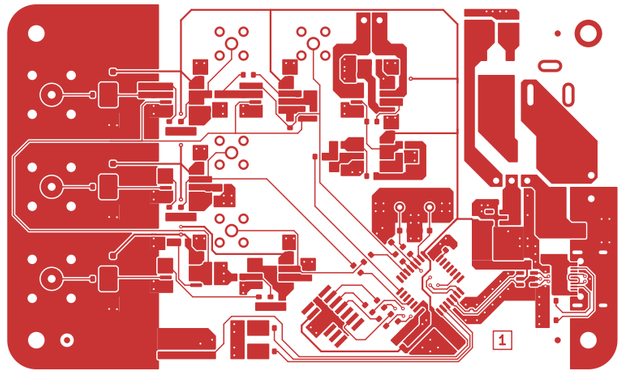
# DREAM FABRICATION DOCUMENT

## L1 (Sig, PWR) (SCALE 1:1)

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| CHK | | | |
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| MFR APP | | | |
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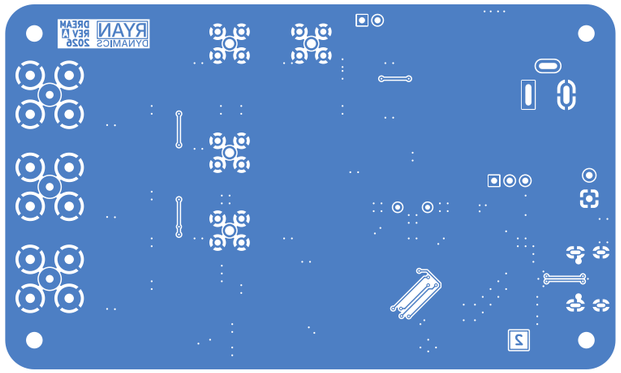
# DREAM FABRICATION DOCUMENT

## L2 (Sig, GND) (SCALE 1:1)



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